HALF YEARLY EXAMINATION

CLASS: IX SUBJECT: GEOGRAPHY TIME: 2 Hrs MAXIMUM MARKS: 90

INSTRUCTIONS:

1. Attempt all questions from **SECTION A**.
2. Attempt any **6 questions** from **SECTION B**.
3. Answer the questions on the externally provided answer sheet.
4. Make sure that all markings on the map are clearly visible and appropriately labeled.
5. Marks are indicated in [ ] .

SECTION A (Attempt All Questions)

Question 1.	**Provide the most appropriate geographical terms or answers for the following:**	**[10]**
	(i) State the average internal temperature of the Earth's mantle. [1]	
	(ii) The geographical term used for new, fertile river deposits. [½]	
	(iii) Name three basic types of igneous rocks. [1 ½]	
	(iv) Name a prominent global location renowned for its basaltic columns. [1]	
	(v) Identify three distinct locations globally where rock salt is naturally found. [1 ½]	
	(vi) A geological phenomenon involving a pressurized water flow erupting from the Earth's interior. [½]	
	(vii) The specific country where Mount Cotopaxi is located. [1]	
	(viii) Name a prominent shield volcano located in Ethiopia. [2]	
	(ix) Identify a modern, reliable scale used for measuring the total energy released by an earthquake, which is considered superior to the Richter Scale. [1]	

Question 2.	**On the provided outline map of the world, accurately locate and label the following:**	**[10]**
	(i) The Velds [2]	
	(ii) Mount Fuji [2]	
	(iii) Senegal River [2]	

	(iv) Lake Titicaca [2]	
	(v) Lake Baikal [2]	

Question 3.	Draw a well-labeled cross-sectional diagram of a waterfall, clearly illustrating the collapse of the hard rock overhang and the geomorphological process of undercutting.	**[3]**

Question 4.	Imagine you are situated on the top floor of your house when a supposedly extinct volcano nearby violently erupts. Narrate the geological sequence of events you would witness and describe the immediate aftermath in your vicinity.	**[2]**

Question 5.	Mount Napier serves as an important geological source of information. What specific insights regarding planetary interior processes does it provide?	**[2]**

Question 6.	*A standard global communication signal is broadcast from a transmission facility located at exactly **74° 30' West** longitude. The signal is sent out at exactly **03:45 AM (Local Time) on April 1, 2024**.*	**[3]**
	*An observer at a receiving station located at exactly **110° 15' East** longitude needs to record the exact arrival time and date.*	
	Calculate the exact local time (using the 24-hour clock) and the precise date (Day, Month, Year) at the receiving station (110° 15' E) at the moment the signal is broadcast. Show all steps of your calculation clearly.	

SECTION B (Attempt Any 6 Questions)

Question 7.	(i) Thoroughly explain the Carbon Cycle and its impact on the Earth. Furthermore, to mathematically and geographically prove the spherical shape of the Earth, describe four distinct phenomenological resources (evidences). Provide a detailed description and a supporting diagram for each of the four evidences. [5]	**[10]**
	(ii) Calculate the exact perpendicular latitudinal distance between Delhi (30° N) and Mumbai (19° N). [2]	

	(iii) If the local time is exactly 8:00 AM on Monday in Sydney (150° E), calculate the exact longitude of a location where the local time is currently 10:00 PM on Sunday. [2]	
	(iv) If the local time is 7:00 AM at 130° W, calculate the longitude of a submerged naval submarine where the local time is currently 3:00 AM. [1]	

Question 8.	(i) State the precise duration of daylight experienced at the following latitudes during the Northern Hemisphere's Winter Solstice: a) 50° N b) 23½° N c) 23½° S d) 50° S e) 66.5° S [5]	**[10]**
	(ii) Explain the literal and geographical meaning of the Latin root word "sistere" in relation to orbital mechanics. [1]	
	(iii) Discuss the comprehensive hypothesis presented by Alfred Wegener and other early geologists regarding the primordial part of earth that once contained all terrestrial life in the known universe, detailing the evidence for its fragmentation. [4]	

Question 9.	(i) List six key structural features and characteristics of fold mountains formed exclusively by lateral tectonic compression. [3]	**[10]**
	(ii) With respect to the North American Cordillera, identify its specific landform type and explain its chronological formation process. Include a subtle, well-labeled diagram. [2]	
	(iii) Compare the distinct types of coastal plains found along the East Coast of India and the Gulf Coast (USA), explicitly outlining their different formation processes. Additionally, explain the foundational geological concept regarding Germany's extending landmass concerning its low geographical position. [3]	
	(iv) Identify the most abundant rock classification on the Earth's surface and explain its dominant presence based on the exogenous agents responsible for its continuous creation. [2]	

Question 10.	(i) Identify the natural parent material (protolith) from which Schist is geologically formed. [1]	**[10]**
	(ii) Suppose you live in the Philippines and discover volcanic 'lapilli' scattered behind your house. Explain the exact geological phenomena responsible for its presence. What other	

	related volcanic ejecta would you hypothetically expect to find nearby? [2]	
	(iii) If you were to embark on a geological expedition to the Canary Islands, explain the rationale for this trip in terms of volcanic structures. What specific features dictate the tectonic belt of its location? [2]	
	(iv) Explain the mechanical propagation of Rayleigh waves and clearly distinguish them from Love waves. Provide a diagram illustrating their movement and discuss their specific surface displacement and potential for destruction. [2]	
	(v) Illustrate and briefly explain the distinct seismic shadow zones created by Primary (push) and Secondary (shear) waves. [1]	
	(vi) Explain how Newton's First Law of Motion (Inertia) is fundamentally applied in the instrumentation used to record the passage of these seismic waves at a specific geographical point. [2]	

Question 11.	(i) Provide one historical example of a major catastrophic tsunami, including its exact date. Additionally, provide three historical examples of significant global earthquakes, detailing their specific dates and epicenter locations. [2]	**[10]**
	(ii) During a field study to the Rann of Kutch, what specific type of intense weathering would you predominantly observe? Provide detailed geographical reasons for your conclusion based on the region's climate. [2]	
	(iii) The Mekong River in Asia forms prominent river cliffs during a specific stage of its fluvial course. Identify this stage and describe two special geomorphological characteristics associated with it. [2]	
	(iv) A naval research vessel departs from Port Alpha, located at exactly **178° 40' East** longitude. The exact Local Mean Time (LMT) of departure is **23:30 (11:30 PM) on February 28, 2024**. The vessel travels continuously eastward until it docks at Port Beta, located at exactly **176° 50' West** longitude. The total travel time for this journey is exactly **26 hours and 45 minutes**. Calculate the exact Local Mean Time (LMT), including the precise time and calendar date, at Port Beta at the exact	

	moment the vessel arrives. Show all mathematical deductions clearly. [4]	

Question 12.	(i) Prove mathematically or via geographic geometry that at 60° latitude in either hemisphere, the total circumference of the latitudinal circle is exactly half the circumference of the Equator. [5]	**[10]**
	(ii) Thoroughly explain the geomorphological and hydrological phenomena that dictate how an oasis is formed and subsequently discovered in a hyper-arid desert mode. Use Diagrams to explain.[3]	
	(iii) Draw a comprehensive, well-labeled global diagram accurately depicting the prevailing directional flow of: 1) Trade Winds 2) Westerlies in both the Northern and Southern Hemispheres. [2]	

Question 13.	**Read The Passage Below and Answer the FOLLOWING QUESTIONS..** **The Global Thermohaline Paradox** Oceanic circulatory systems do not merely transport fluid; they are the primary architects of planetary atmospheric thermodynamics, biological distribution, and commercial maritime viability. Analyze the following global anomalies and provide exhaustive, geographically sound justifications for each phenomenon.	**[10]**
	(A) The Precipitation-Desiccation Dichotomy [2] An exploratory vessel sailing along the western boundary of a southern hemisphere landmass observes a hyper-arid, barren landscape stretching far inland, which owes its existence directly to a specific equator-bound hydrodynamic flow. Conversely, a massive poleward-moving "western boundary current" in the northern Pacific (often characterized by its deep, dark waters) sustains immense moisture-holding capacity in the overlying air masses, bringing heavy precipitation to adjacent coasts. Elucidate the exact contrasting thermodynamic interactions between the sea-surface temperatures and the overlying atmosphere that enforce strict desertification in the former region while ensuring abundant coastal rainfall in the latter.	

	(B) The Latitudinal Freezing Anomaly [3] Port Alpha is geographically classified as the most equatorward port globally to suffer severe seasonal freezing, requiring heavy ice-breaking maritime vessels for nine months of the year. Curiously, a neighboring marginal sea avoids this deep freeze entirely due to ten-meter tidal fluctuations. Meanwhile, Port Beta (e.g., Spitsbergen), located astronomically deeper into the frigid northern latitudes, remains completely unfettered by winter ice, maintaining year-round open commerce. Identify the specific subarctic and tropical oceanic drifts dictating the freezing/thawing fate of both ports. Detail the exact collision dynamics off the eastern shore of an Asian archipelago that birth the circulatory system punishing Port Alpha, and explain why this specific cold-water collision zone paradoxically supports a massive concentration of bio-nutrients.	
	(C) The Trans-Atlantic Navigational Hazard Zone [3] A commercial freighter navigating a notorious convergence zone off the eastern coast of North America encounters a catastrophic drop in visibility, immediately followed by structural damage from a massive floating freshwater glacial fragment. Furthermore, this exact oceanic intermingling zone is the genesis point for violent, destructive atmospheric cyclonic systems that track along the coastline. Deduce the exact state-change mechanism (thermodynamic condensation) causing this blinding visibility, specifying the thermal origins of the two converging oceanic streams. Trace the geographic origin of the glacial fragment and identify the specific carrier stream responsible for transporting it into trans-Atlantic shipping lanes. Finally, explain the meteorological instability that breeds violent hurricanes exactly where these two specific fluid bodies mingle.	
	(D) The Benthic Wealth Multiplier [2] The very same treacherous, fog-bound convergence zone from part (C) is paradoxically an area of immense economic extraction, mirroring a mathematically similar collision zone in the North Pacific. Describe the complete biological cascade—beginning with the movement of organic material and microscopic entities, up to higher trophic levels—that transforms these dangerous, stormy meeting points into the highest concentration of commercial marine life on Earth.	